BBQ (Better Binary Quantisation)

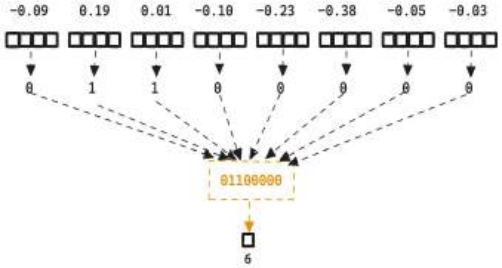
for Inference Infrastructure Optimisation

×

Arkadip Basu

Principal Engineer
AI & Platforms Engineering

Disclaimers



Nature of Content

The information, content, and materials provided in this presentation are for educational and informational purposes only. The speaker has compiled this material based on general industry knowledge, personal expertise, and publicly available data.

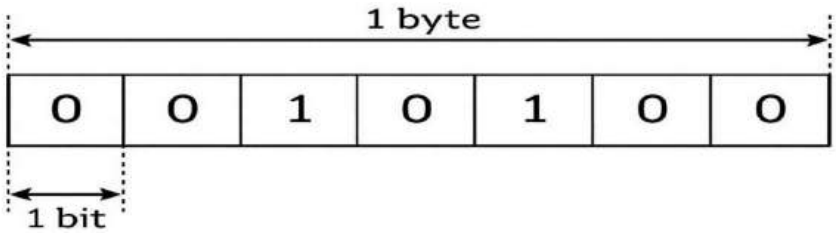
Public Domain & Sources

- **Media & Images:** All images, graphics, and visual media used in these slides are sourced from the public domain and are used under creative fair use guidelines.
- **Articles & Research:** Any technical articles, case studies, or frameworks referenced are available on public media or via open-source platforms.
- **Attribution:** Every effort has been made to provide proper attribution to the original creators where applicable.

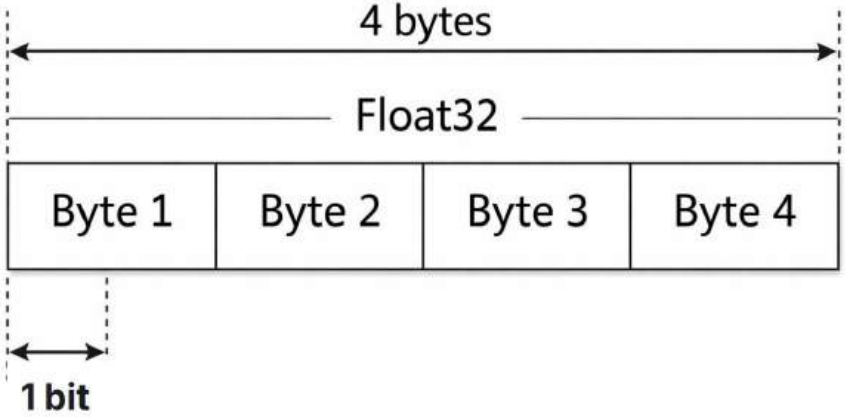
Non-Representation of Organization

- **Independent Views:** The views, thoughts, and opinions expressed in this presentation belong solely to the speaker and do not necessarily reflect the official policy or position of their employer or parent organization.
- **Confidentiality:** This presentation does not contain any confidential, proprietary, or trade-secret information belonging to the speaker's parent organization. No internal projects, private data, or non-public intellectual property have been disclosed.

How many Bits are in 1-byte



How many Bytes are in 1-float32 vector



How many Bytes does a float32 Image 1920 x 1080 pixels need ?

(1920 x 1080) x 4 bytes = 8294400 bytes. If it is RGB image it requires (8294400 x 3) bytes = 24.88 MegaBytes (in raw)

Camera Raw 9.9 - Canon EOS 6D

R: ---
G: ---
B: ---

f/7.1 1/100 s
ISO 100 24-105@24 mm

Basic

White Balance: As Shot

Temperature 5700

Tint +8

Auto Default

Exposure +0.90

Contrast 0

Highlights -100

Shadows +100

Whites 0

Blacks 0

Clarity +40

Vibrance +50

Saturation 0

30.7% NYC Sunset.cr2

Save Image... Adobe RGB (1998); 8 bit; 5472 by 3648 (20.0MP); 300 ppi Open Image Cancel Done

Preview Publish

4U8A7733.JPG

Metadata Keywords Filter

f/1.4 1/250 ISO 3200

1365 x 2048 733 KB 240 ppi sRGB RGB

IPTC Core

IPTC Extension

Camera Data (Exif)

Exposure Mode	Manual
Focal Length	85.0 mm
Lens	SAMYANG RF 85mm F1.4
Date Time Original	12/23/2020, 3:44:08 PM
Flash	Did not fire
Metering Mode	Evaluative
White Balance	Auto
Model	Canon EOS R6

In the era of Gen-AI,

How we handle operations on images that too is big in size

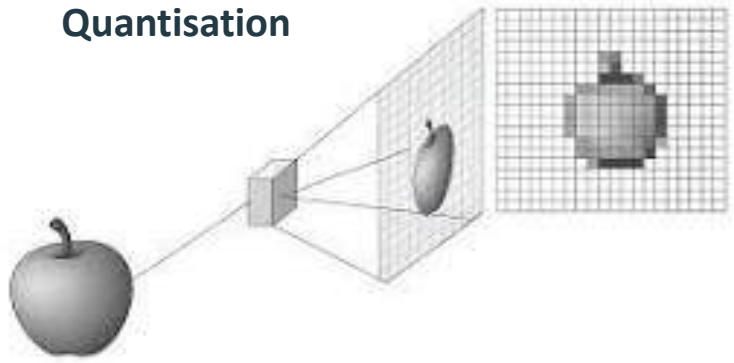
Is it Compression or Quantisation

IND 273-4 P3 10 MIN 4 BALLS 11 SL 274-6 (50)
PARTNERSHIP 48 RUNS FROM 41 BALLS

Everything that we/system Generate / Create/ Do / Process, leaves footprint & that is saved as **Data in Disk**

Feature	Data Compression	Data Quantisation
Core Definition	The process of encoding information using fewer bits than the original representation.	The process of mapping a large set of input values to a smaller, discrete set of output levels.
Data Retention	Can be **Lossless** (bit-perfect recovery) or **Lossy** (approximate recovery).	Inherently **Lossy**; it intentionally discards precision to simplify data.
Hardware Impact	Primarily reduces **storage size** and bandwidth requirements for data transfer.	Primarily increases **computation speed** by enabling low-precision integer arithmetic.
Primary Technique	Uses statistical redundancies (e.g., Huffman coding, Run-Length Encoding).	Uses "binning" or rounding (e.g., converting 32-bit floats to 8-bit integers).
Common Application	General file storage (ZIP), image formats (PNG/JPEG), and video streaming (H.264).	Optimizing Deep Learning models (LLMs) to run on mobile phones and edge devices.

Quantisation



Reducing Precision: Quantization reduces the precision of the data by representing it using fewer bits or numbers.

Discrete Representation: Continuous values are mapped to a finite set of discrete values, called quantization levels.

Loss of Information: Quantization is a lossy process, meaning that some information is lost during the conversion.

Applications

Signal Processing: Quantization is used in signal processing to convert analog signals to digital signals.

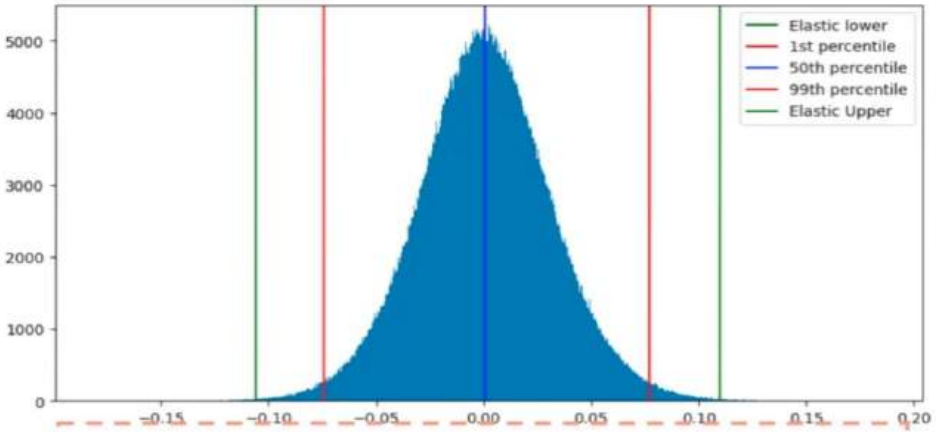
Image and Video Compression: Quantization is used in image and video compression algorithms, ex: JPEG & MPEG.

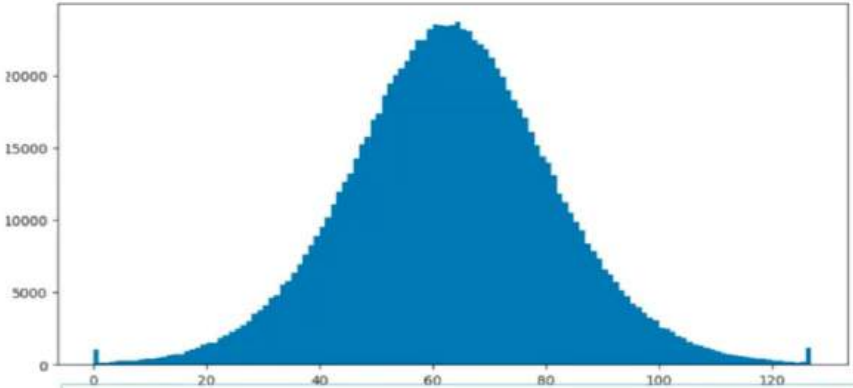
Machine Learning: Quantization is used in machine learning to reduce the precision of model weights and activations, making models more efficient and scalable.

Benefits:

Reduced Storage, Improved Computational Efficiency, Faster Inference

Quantisation to the rescue





[1.0, -0.28, ..., 3.14, 0.42]

min/max normalize w/ quantiles

[8, -4, ..., 120, 5]

4x - 8x reduction in space

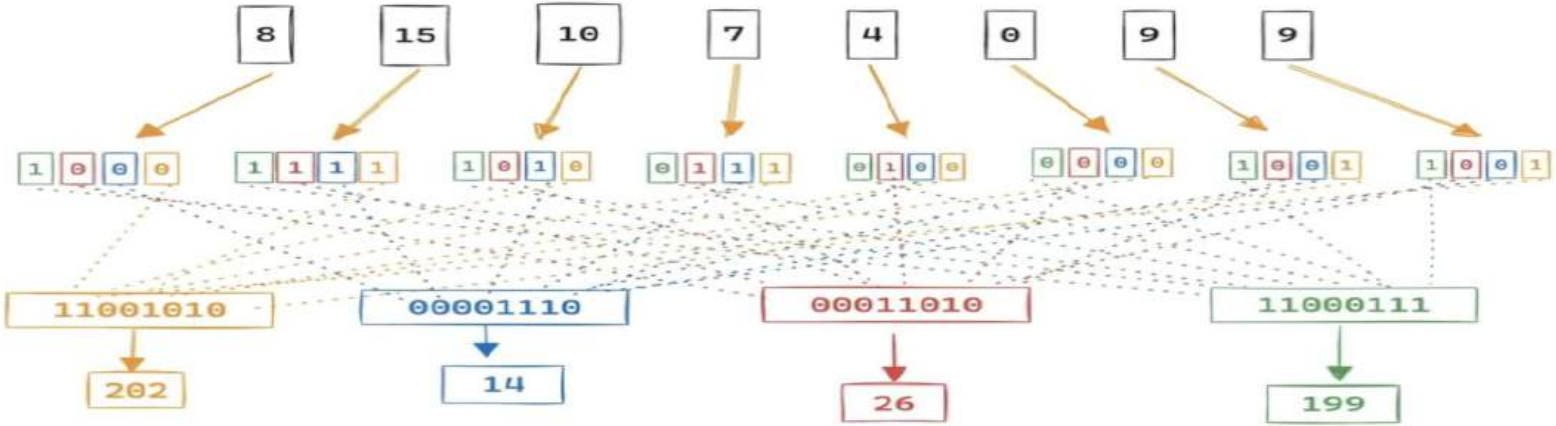
Almost zero accuracy loss

faster search

Perfect Fit for Lucene

elastic

https://www.elastic.co/search-labs/blog/int4-scalar-quantization-in-lucene


8
15
10
7
4
0
9
9
1000
1111
1010
0111
0100
0000
1001
1001
11001010
00001110
00011010
11000111
202
14
26
199

Challenges in Modern Vector Search

Sample calculation: What is the overall latency of retrieving 30 256kB images from one server?

Latency Numbers Everyone Should Know

Operation	Time in ns	Time in ms (1ms = 1,000,000 ns)
L1 cache reference	1	
Branch misprediction	3	
L2 cache reference	4	
Mutex lock/unlock	17	
Main memory reference	100	
Compress 1 kB with Zippy	2,000	0.002
Read 1 MB sequentially from memory	10,000	0.010
Send 2 kB over 10 Gbps network	1,600	0.0016
SSD 4kB Random Read	20,000	0.020
Read 1 MB sequentially from SSD	1,000,000	1
Round trip within same datacenter	500,000	0.5
Read 1 MB sequentially from disk	5,000,000	5
Read 1 MB sequentially from 1Gbps network	10,000,000	10
Disk seek	10,000,000	10
TCP packet round trip between continents	150,000,000	150

Therefore, it is possible to read:

- sequentially from HDD at a rate of ~200MB per second
- sequentially from SSD at a rate of ~1 GB per second
- sequentially from main memory at a rate of ~100GB per second (burst rate)
- sequentially from 10Gbps Ethernet at a rate of ~1000MB per second

https://static.googleusercontent.com/media/sre.google/en//static/pdf/rule-of-thumb-latency-numbers-letter.pdf

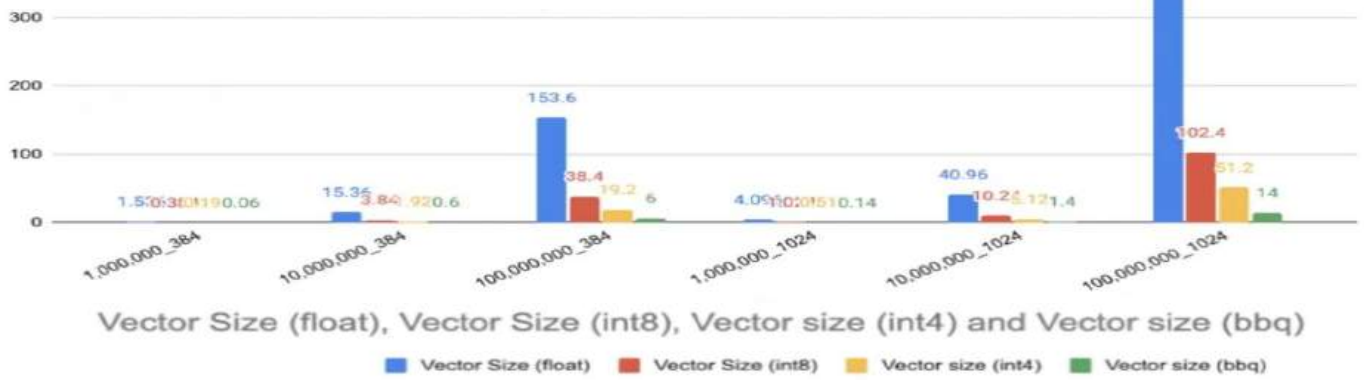
How Better Binary Quantization is done

- Better Binary Quantization (BBQ) is a lossy compression technique for vector embeddings that encodes real-valued vectors into compact binary strings.
- Better Binary Quantization compresses data by converting numbers into 0s and 1s.
- BBQ uses multiple thresholds which are typically optimized offline to minimize the quantization error, often using a greedy algorithm. The result is a highly compressed binary representation that significantly reduces storage and memory bandwidth requirements.
- This compression introduces approximation error, but it is designed to preserve the relative distances between vectors, BBQ is highly effective for Approximate Nearest Neighbor (ANN) search, as the binary codes maintain the essential structure needed for efficient similarity search.

Memory to Storage ratio for better performance

Around 1:30 memory-to-disk ratio (e.g., 64GB RAM to 1.9TB Disk). [based on discussion of Elastic v6.x, 2001]
AI/GPU Training: 3:2 to 2:1 ratio of system RAM to GPU memory is for handling large models.

Performance Impact: RAM provides very low latency for immediate tasks, while storage holds data for the long term. Upgrading RAM is generally more effective for speeding up multitasking, while upgrading storage increases capacity. The "Enough" Principle: RAM only improves performance if you are currently running out. Adding more RAM once you have sufficient amount does not improve speed.

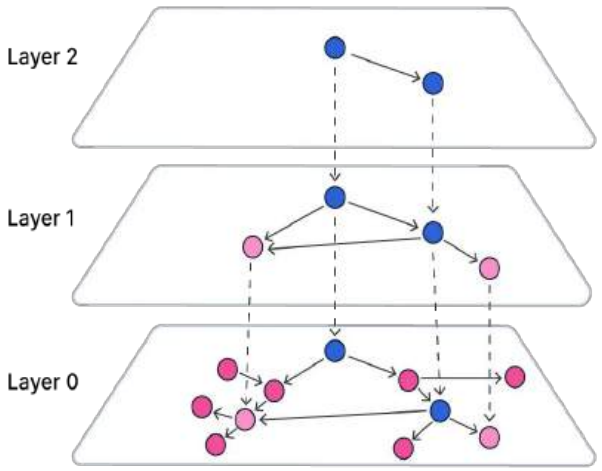


https://discuss.elastic.co/t/clarification-about-recommended-memory-disk-ratio-of-1-30/221547

Bright cyan, Black, White, mild Gray

Even though it is virtual or distorted or corped, we still get the meaning.
Owing to our Pre-trained memory. Same goes for digital, leverage and execute prior knowledge. Representation is important

Hierarchical Navigable Small World



- HNSW is an approximate nearest neighbor (ANN) search algorithm that organizes data points into a multi-layer graph.
- Each layer is a proximity graph where nodes connect to nearby neighbors, with higher layers being sparser.
- Queries start at the top layer and greedily navigate down to closer nodes layer by layer.
- It achieves fast search (logarithmic complexity) with high recall by balancing exploration and exploitation.
- Widely used in vector databases for similarity search in embeddings (e.g., NLP, recommender systems).

https://www.elastic.co/search-labs/blog/hnsw-graph

Elastic BBQ vs HNSW: A Comparison

Elastic BBQ (also known as DiskANN) and HNSW are both popular indexing algorithms for approximate nearest neighbor (ANN) search. While both algorithms are designed for efficient similarity search, they have different strengths and weaknesses.

Advantages of Elastic BBQ over HNSW:

Disk-based Indexing: Elastic BBQ is designed to work efficiently on disk-based storage, making it suitable for large datasets that don't fit in memory. HNSW, on the other hand, is typically used in-memory.
Memory Efficiency: Elastic BBQ can be more memory-efficient than HNSW, especially for large datasets, since it doesn't require loading the entire index into memory.
Support for Dynamic Data: Elastic BBQ is designed to handle dynamic data, allowing for efficient insertion, deletion, and update operations. HNSW, while capable of handling some level of dynamism, is generally more suited for static or semi-static datasets.
Query Performance: Elastic BBQ can offer competitive query performance to HNSW, especially for larger datasets, due to its optimized disk I/O and caching strategies.

When to Choose Elastic BBQ over HNSW:

Large Disk-based Datasets: When working with massive datasets that don't fit in memory, Elastic BBQ's disk-based indexing can be a better choice.
Dynamic Data: When the dataset is constantly changing, Elastic BBQ's support for dynamic data can be beneficial.
Memory-constrained Environments: Areas with limited memory resources, Elastic BBQ's memory efficiency is advantageous.

Infrastructure Savings at Scale | High Throughput AI Ops

Memory Footprint Breakdown, for 100M Vectors, 384 Dims:

- **Raw (float32):** 143.05 GB
- **Int8 Scalar:** 36.14 GB
- **BBQ (1-bit): 5.77 GB**
- **Indexing Speed:** 20-30x faster than traditional Product Quantization (PQ).
- **Merging:** BBQ leverages existing centroids during segment merges for efficient weighted-average updates.
- **Query Latency:** 2-5x faster than older quantization methods
- Frictionless Deployment via API.

```
PUT /enterprise-bbq-index
{
  "mappings": {
    "properties": {
      "product_vector": {
        "type": "dense_vector",
        "dims": 384,
        "index": true,
        "index_options": {
          "type": "bbq_hnsw"
        } } } } }
```

```
POST
/enterprise-bbq-index/_search
{
  "knn": {
    "field": "product_vector",
    "query_vector": [ ... ],
    "k": 10,
    "num_candidates": 100
  },
  "rescore": {
    "rescore_vector": {
      "rescore_query": { ... }
    } } }
```

The "Win-Win" Strategy

Don't settle for smaller models (`e5-small`) to save RAM. Use a **powerful model (**e5-large**) with BBQ**.

Why: Get better ranking from large model, lower latency from BBQ bit-scans, & lower costs ~32x memory reduction.

Best Practices & Use case checklist

BBQ is optimized for **384+ dimensions**; lower dimensions may see higher relative overhead from corrective factors.

Centroids update incrementally; reindex to apply BBQ to existing vectors. Also known as Segment Merging

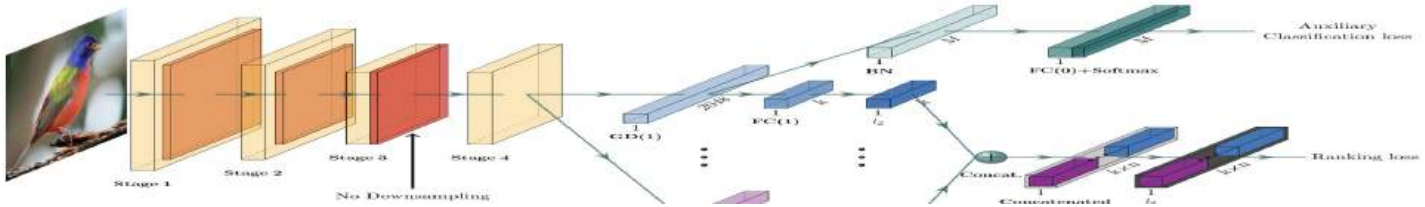
Ensure ML nodes have sufficient RAM (minimum 4GB) to load larger models

Elastic ACORN as compatible partner

Elasticsearch ACORN is used for graph-based indexing algorithm to optimize similarity search over dense vector data. It constructs a graph index that facilitates fast traversal and retrieval of similar vectors. The algorithm is designed to enable efficient and scalable nearest neighbor search, making it suitable for applications with large vector datasets. This also ensures high search performance and accuracy.

Reference Materials

- Elastic BBQ: Better Binary Quantization in Lucene & Elasticsearch
- Better Binary Quantization (BBQ): How & why to implement - Elasticsearch Labs
- Better Binary Quantization (BBQ) | Elasticsearch Reference
- Elasticsearch now with BBQ by default & ACORN for filtered vector search
- Scalar quantization 101: The basics, benefits and applications - Elasticsearch Labs
- https://arkadipbasu.medium.com/bbq-in-elastic-acb1a963e9af
- https://www.youtube.com/watch?v=04NzMt2Nigc
- https://github.com/elastic/elasticsearch-labs/tree/main/supporting-blog-content/how-and-why-bbq
- Better Binary Quantization (BBQ) | Elasticsearch Reference
- https://www.elastic.co/docs/reference/elasticsearch/mapping-reference/bbq#bbq-disk



Arkadip Basu

- Principal Engineer at FinTech Enterprise, applying Resiliency & Scale with AI & ML for business critical services & Emerging Technologies.
- Expertise in developing reliable platform, capable of handling large volume of high frequency transactions across multiple geographies.
- Engaged in Pro-Active algorithm building for efficient Op-Ex planning, analysis & cost optimization. Manager of multiple AI-Agents, Bots.
- IEEE fellow, associated with IIT-Madras, AI4Bharat, Outlier, iDEX.gov
- Masters in AI from IISc Bangalore, specialization in Computer Vision.

Connect on Linkedin

Thank you

for making it so interactive

© https://arkadipbasu.github.io